

Dinesh Kannan

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EDUCATION

North Carolina State University
Masters of Computer Science

August 2023 - May 2025
GPA: 3.85/4.0

Coursework: Automated Software Engineering, Cloud Computing, Neural Networks, Object Oriented Design and Development, Database Management Systems, Design and Analysis of Algorithms

WORK EXPERIENCE

North Carolina State University | Software Developer

August 2025 - Present

- Built full-stack geospatial platform with **React, Node.js, PostgreSQL** serving 100+ students, implementing a custom Timeline Builder with before/after image comparison sliders that increased student research engagement by **35%**.
- Implemented **RBAC authentication** system with **JWT** and role-based workflows, reducing instructor approval time by **40%**.
- Reduced page load times by 75% (3.2s → 0.8s) as measured across 100+ student sessions, by migrating media infrastructure to Cloudinary CDN and offloading 60% of database storage operations.
- Designed automated draft-to-publish pipeline handling 100+ timeline submissions in a semester with **99% accuracy**.

North Carolina State University | Research Assistant

July 2024 - April 2025

- Developed **AI teaching assistant** serving 500+ students with **99.2% uptime** and 1.2s avg response time using **React, Flask, MongoDB and AWS**.
- Built **RESTful API** with FastAPI handling 10,000+ monthly queries, implementing JWT auth and **performance optimization** strategies including database indexing and query caching for sub-second response times.
- Engineered **RAG-based chatbot** using **OpenAI, LangChain**, improving accessibility by **35%** through context-aware responses.
- Automated **CI/CD pipeline** with **GitHub Actions**, reducing deployment time by **40%** and eliminating manual release errors.
- Integrated **AWS Cloud Storage** for scalable media management, handling 50GB+ of course materials with **99.9% availability**.
- Designed **database schema optimization** strategy reducing query latency by **55%** through indexing and query restructuring.

Deloitte | Consulting Analyst

January 2023 - July 2023

- Developed **.NET plugins** for **Dynamics 365**, reducing query response time by **40%** (8s → 4.8s) for 10,000+ record datasets.
- Accelerated **feature delivery by 20%** by leading daily stand-ups and refining backlog grooming, successfully **shipping 10 core features across 6 sprints** with zero post-deployment defects.
- Identified critical vegetation growth patterns by **analyzing 50GB+ of datasets** using Python & SQL, directly influencing client resource allocation strategy and reducing projected maintenance costs by 15%.
- Built custom business automation workflows with Power Automate and Azure Data Factory, streamlining cross-system data processing for 5 client projects and reducing manual effort by 50 hours/month.

TECHNICAL SKILLS

Languages & Frameworks: Python, JavaScript/TypeScript, React, Node.js, Flask, FastAPI, HTML/CSS, .NET

Cloud & DevOps: AWS (Lambda, S3, EC2), Google Cloud Platform, Docker, CI/CD (GitHub Actions), Distributed Systems

Databases & Tools: PostgreSQL, MongoDB, MySQL, Git, Agile/Scrum, Jest/Pytest, Postman

AI & ML : LLMs, Agentic AI, PyTorch, RAG, OpenAI, LangChain, Transformers, Pandas, Scikit-learn

PROJECTS

Real-Time YouTube Sentiment Analysis via Chrome Extension | Python, AWS, React, Docker

- Built full-stack app processing **50,000+ YouTube comments** in real-time using containerized microservices on **AWS ECS**.
- Designed **RESTful API with FastAPI** and integrated **AWS Lambda** for serverless processing achieving **200ms average latency**.
- Implemented **Docker-based deployment** pipeline with **99.5% uptime** across 3 AWS availability zones.
- Optimized comment processing pipeline using **hash maps and priority queues**, reducing memory usage by **40%** and enabling real-time analysis of 50,000+ comments with $O(n \log n)$ complexity.

Ellipsis: Podcast Generation Platform | Python, Flask, React, Redis, Orpheus TTS

- Automated podcast generation achieving 95% natural-sounding speech quality as measured by user feedback surveys, by integrating **Perplexity's deep research API with Orpheus TTS** to produce multi-topic content in under 3 minutes per episode.
- Built **microservice architecture with FastAPI** handling 5 services: research, scripting, TTS, storage, and API gateway.
- Implemented **Redis caching** layer reducing API costs by **60%** and improving response times to **<500ms**.

BurnoutExtreme | Flask, React.js, MongoDB, Google Maps API

- Increased user retention by 27% as measured over 8-week study period, by building modular MongoDB-based diet tracking system with personalized goal-setting features for 200+ daily active users.
- Achieved **92% test coverage** using **pytest, TDD**, reducing production bugs by **78%** for 200+ daily active users.